

Wing Velocity Control System for Testing Body Motion Control Methods for Flapping Wing MAVs

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This work is focused on testing the viability of split-cycle constant-period frequency modulation for controlling a flapping wing micro air vehicle. By varying the velocities of the upstroke and downstroke portions of a flapping wings' motion, non-zero cycle-averaged drag can be generated. This results in the ability to control the roll and horizontal translation of the vehicle by varying the split-cycle parameter. In this paper, a flapping wing prototype is developed for testing the split-cycle technique. A fuselage, wings, motor control electronics, and motor control software are created to allow wireless control of the motion of the flapping wing vehicle. The complete vehicle is then placed on an air table for experimental testing. The air table provides the lift equal to weight condition and constrains the pitch and yaw degrees of freedom while allowing for roll rotation and horizontal translation. In this way, testing of the degrees of freedom which can be manipulated via split-cycle, namely roll and horizontal translation, can be performed. The air table tests reveal that split-cycle can be used to control these two degrees of freedom.

I. Introduction

A prototype flapping wing aircraft, whose motion was wirelessly controlled without using servos to manipulate the wing kinematics or conventional aerodynamic surfaces such as elevons, ailerons, or rudders, was created. Two brushless direct current (BLDC) motors were used to manipulate the motion of flapping wings such that differential upstroke and downstroke velocities can be generated over the course of a wingbeat cycle. The system was tested on an air table and demonstrated the ability to control two degrees of freedom, namely, roll and horizontal translation. The air table experiment qualitatively validates the theory that split-cycle constant-period frequency modulation is a viable method for controlling flapping wing micro air vehicles (MAVs).¹ More specifically, this experiment demonstrated that temporal asymmetries in wing strokes over a wingbeat cycle can be used to manipulate the cycle-averaged horizontal force and rolling moment acting on a flapping wing vehicle. When implemented on a flight weight system, these same two BLDC motors will also allow the control of two additional degrees of freedom, namely, yaw and vertical translation, as demonstrated by past experiments by Wood² and Anderson.³ To the knowledge of the authors, the system presented in this work represents the first aerospace vehicle that has the ability to directly control more degrees of freedom than the number of physical actuators present on the vehicle. Such

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an aircraft configuration offers benefits over previous designs that are encumbered with the additional weight of servos, empennage effectors, and/or complex servo driven mechanisms that manipulate wing kinematics. The weight and complexity associated with previous systems has effectively been replaced by a simple drive mechanism that is controlled by relatively complex software that has been carefully written to run in real-time on surface mount microcontrollers with modest processing capabilities.

Wingbeat velocity shaping has been proposed as a method for controlling the body motion of flapping wing MAVs.^{1,4} Simulation results indicated that it is possible to control four degrees of body freedom using two physical actuators that independently drive the stroke plane motion of each wing. The simulation results of past work were based upon blade element theory and as such contained numerous assumptions:

1. Rigid wings with zero mass
2. Perfect tracking of wing actuator commands
3. Wing angle of attack passively regulated by wing rotation stops
4. Wing transitioned between stops instantaneously at stroke reversal points

The degree to which these assumptions are justified on flight-weight systems has been partially evaluated in bench test studies.^{3,5} Anderson³ provided experimental evidence that showed that the cycle-averaged blade element model gave reasonably good predictions of measured cycle-averaged forces and moments in the vertical, horizontal, pitch, roll, and yaw axes. Unfortunately, the force balance used to take the measurements had a low signal to noise ratio when measuring the horizontal force and rolling moment. The magnitude of these variables was so small that they were near the noise threshold of the balance and while the statistics showed that the cycle-averaged blade element theory correctly predicted the trends, the variance of the estimates was high. Force balances produce signals that are based upon displacement under the influence of a force or moment. Balances with high bandwidth tend to be stiff and balances with high spatial or force resolution tend to be low stiffness. Thus, it is difficult to obtain high-bandwidth and high resolution force measurements from MAVs because of this inherent physical limitation.

The use of an air table is proposed to evaluate the efficacy of wing velocity modulation based on the split-cycle constant-period frequency modulation¹ body control strategy. This method takes advantage of the fact that the wingbeat frequencies are separated from the rigid body dynamics of the MAV. Past work hypothesized that the aerodynamic and inertial forces associated with the flapping wings could be cycle-averaged to produce a model of the system that would be sufficient to predict the controlled motion of the body of the aircraft. The nearly frictionless surface of an air table provides an environment that can be used to test the ability of the wing stroke velocity modulation methods to produce desired motion in the roll axis and horizontal direction as an intermediate step towards free flight.

A major technical hurdle of testing wing velocity control strategies on an air table is that wireless control of the wing motion is required. Direct control of wing velocity is not achievable with conventional radio controlled (RC) wireless servo control because RC servos have insufficient bandwidth and power to achieve the wing velocities required for flight. The two primary candidates for flapping wing MAV prime movers are bimorph piezo electric elements and direct current (DC) motors. Piezo elements require a high voltage-low current power source which is not readily available in the size, weight, and power range needed to support free flight or un-tethered air table experiments. Therefore, BLDC motors were chosen as the prime movers. An approach to BLDC motor-based wireless wing velocity control is presented in this manuscript. Two BLDC motors are used to independently control the stroke velocity of two flapping wings. The motor assembly contains position sensors and an integrated planetary gear box. A four-bar linkage transforms the rotary motion of the motor into the flapping motion of the wing.

Figure 1 shows a block diagram of the system used to control the motion of the flapping wings. A radio frequency (RF) transmitter and receiver are used to convey user commands to the vehicle. Three microcontrollers (MCUs) are used to convert receiver signals into motor commands and drive the BLDC motors. Current amplifiers are wired directly to the phases of each BLDC motor. Motor sensors include encoders, for position and velocity feedback control, and Hall effect sensors, for commutation. More detail on motor control will be provided in the following sections.

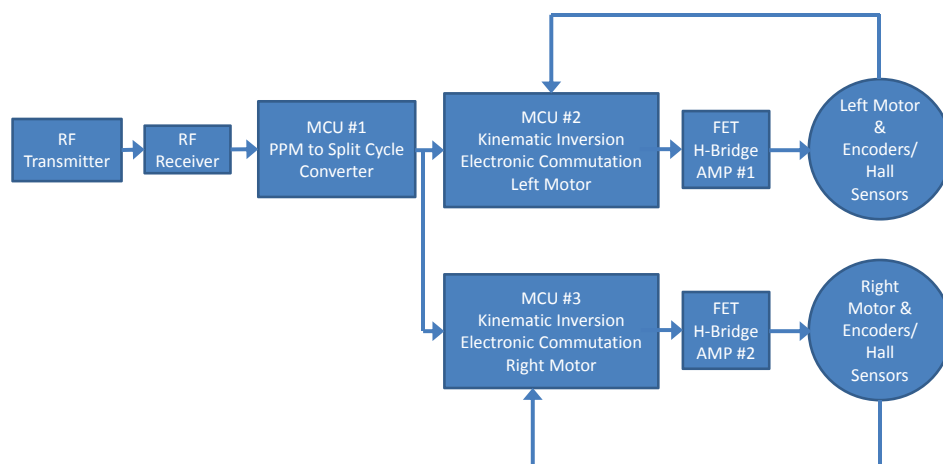


Figure 1: Conceptual Block Diagram of Wireless Wing Velocity Control System.

II. Split-Cycle Constant-Period Frequency Modulation

Figure 2 shows a computer aided design (CAD) model of the flapping wing vehicle with the body-axis coordinate system (x_b, y_b, z_b) . Figure 2a shows a top view of the flapping assembly and Figure 2b shows an isometric view. The parts of the figure stretching out along the z_b axis are the connecting wires for the BLDC motors and encoders. The x_b axis is defined as up, the z_b axis points forward, and the y_b axis points out the right hand side of the vehicle. The wings flap in the y_b, z_b plane. Zero wing position is defined as

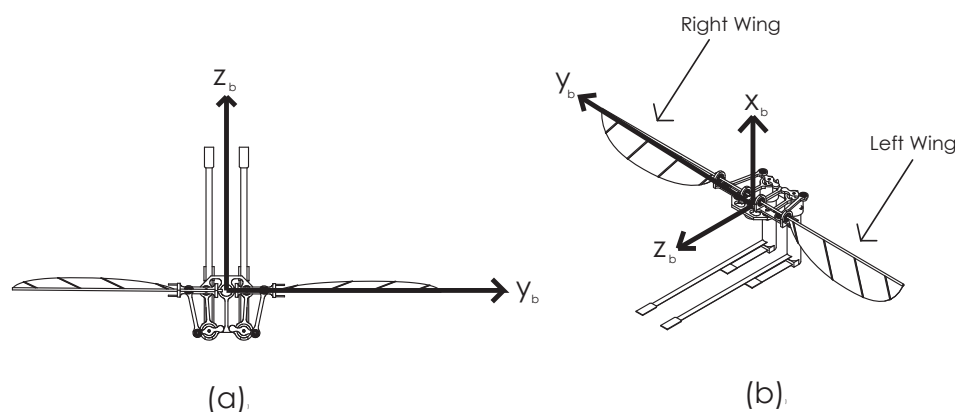


Figure 2: Flapping Wing Vehicle.

the right wing coincident with the y_b axis and the left wing coincident with the $-y_b$ axis. The stroke plane, which denotes the plane in which the wing spar lies throughout each wingbeat, lies in the x_b, y_b plane. From a top down view (Figure 2a), the right wing's positive rotational direction is defined as counter-clockwise. In the same view, the left wing's positive rotational direction is defined as clockwise. For a wing starting in the foremost position, that is, its most positive wing position excursion, a complete wingbeat cycle is defined as an upstroke followed by a downstroke. The upstroke is defined as starting at the wing's maximum positive position (forward of the origin of the body-axes) and moving to its most negative position (aft of the origin of the body-axes), while the downstroke is defined as starting at the wing's most negative position and moving to its most positive position. The extrema of the wing position are ± 1 rad and are dictated by the four-bar linkage mechanism.

The fundamental idea of split-cycle constant-period frequency modulation (SCCPFM) is to, for a given

period, piece together two half cosine waves such that the period remains unaltered while the half cycle durations may differ from each other. When no split-cycle modulation is applied, the two half cosine waves have the same frequency, namely that corresponding to the given period. When non-zero split-cycle modulation is applied, either the upstroke or the downstroke will be impeded, while the opposite stroke is advanced. The development of the split-cycle parameters for the right wing is exactly the same as for the left wing. Therefore, only general expressions will be considered.

For symmetric flapping, which means the upstroke and downstroke utilize the same frequencies, the wing position is defined as

$$\begin{aligned}\phi_U &= \cos(\omega t), & 0 < t < \frac{\pi}{\omega} \\ \phi_D &= \cos(\omega t), & \frac{\pi}{\omega} \leq t < \frac{2\pi}{\omega}\end{aligned}\quad (1)$$

where ϕ_U, ϕ_D are the upstroke and downstroke wing positions, respectively, for either the right or left wing and ω is the symmetric or fundamental wingbeat frequency. When flapping symmetrically, the magnitude of the wing velocities on the upstroke and downstroke are equal.

With SCCPFM, for the first half of the cycle (the upstroke), the frequency is altered from ω to $\omega - \delta$, while for the second half of the cycle (the downstroke), the frequency is altered from ω to $\omega + \sigma$. Thus, the position of the wing is defined by

$$\begin{aligned}\phi_U &= \cos[(\omega - \delta)t], & 0 < t < \frac{\pi}{\omega - \delta} \\ \phi_D &= \cos[(\omega + \sigma)t + \xi], & \frac{\pi}{\omega - \delta} \leq t < \frac{2\pi}{\omega}\end{aligned}\quad (2)$$

where δ is the split-cycle parameter, $\sigma = \frac{\delta\omega}{\omega - 2\delta}$ is the frequency modification parameter, and $\xi = \frac{-2\pi\delta}{\omega - 2\delta}$. If $\delta = 0$, the fundamental wingbeat frequency for both the upstroke and downstroke is achieved and both strokes use the same frequency. For $\delta > 0$, the upstroke has a lower frequency, as compared to the fundamental frequency, and is impeded, while the downstroke has a higher frequency and is advanced. For $\delta < 0$, the upstroke has a higher frequency, as compared to the fundamental frequency, and is advanced, while the downstroke has a lower frequency and is impeded. By altering the frequencies of the upstroke and downstroke, non-zero cycle-averaged drag can be generated, resulting in the ability to translate fore, translate aft, and roll the vehicle. A nonzero split-cycle parameter yields a difference between the velocities of the upstroke and downstroke. This velocity difference results in different dynamic pressure on the wings and hence, different aerodynamic forces generated by the flapping wing.

Figures 3 and 4 show wingbeat patterns for $\delta > 0$ and $\delta < 0$ along with a symmetric wingbeat. In Figure 3, the upstroke is impeded and the downstroke is advanced, while in Figure 4, the upstroke is advanced and the downstroke is impeded. Figures 5 and 6 show the wing velocities for both positive and negative values of split-cycle parameter. The impeded stroke has a much lower velocity than the advanced stroke, resulting in lower dynamic pressure on the wing and a reduction in aerodynamic forces produced by wing flapping. The split-cycle parameter, δ , along with ω for the right wing and left wing, are control parameters used to generate forces and moments on the vehicle.

Figures 7 and 8 show how different degrees of freedom of body motion are controlled via the symmetric and asymmetric frequencies. In each figure, long arrows denote a high velocity stroke, while short arrows denote a low velocity stroke. In Figure 7a, both wings flap with the same frequency. Flapping faster generates more lift and flapping slower less lift, hence, the vertical degree of freedom can be controlled by symmetric frequency. In Figure 7b, the yaw rotation is controlled. Here, one wings flaps fast on both the upstroke and downstroke, while the other flaps slower on both strokes. The wing flapping faster generates more lift than the wing flapping slower, resulting in a rotation about the z_b axis (yaw). In Figure 8a, the roll rotation is controlled. Here, one wing flaps fast on the upstroke and slow on the downstroke, while the other wing does the opposite, that is, it flaps slow on the upstroke and fast on the downstroke. The result is a nonzero drag couple that rotates the vehicle about the x_b axis. Lastly, Figure 8b shows the horizontal translation degree of freedom. In this case, both wings experience high velocity upstrokes and both wings experience low velocity downstrokes. The net effect is a nonzero cycle-averaged drag pointing in the z_b direction which allows the vehicle to translate forward. By making both upstrokes low velocity and both downstrokes high velocity, backward translation in the $-z_b$ direction can be accomplished.

For the experiment considered in this work, the air table constrained the degrees of freedom shown in Figure 7. More specifically, the air table generated sufficient lift to float the flapping assembly on the table. The flapping wings produced additional lift, however, the flapping frequency was not high enough to generate enough lift to raise the flapping assembly off the air table. The flapping assembly was mounted to a puck via

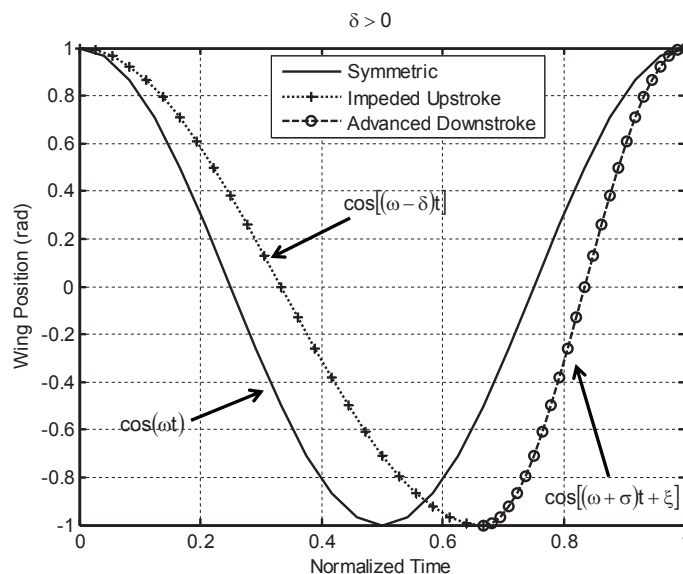


Figure 3: Wing Position for $\delta > 0$.

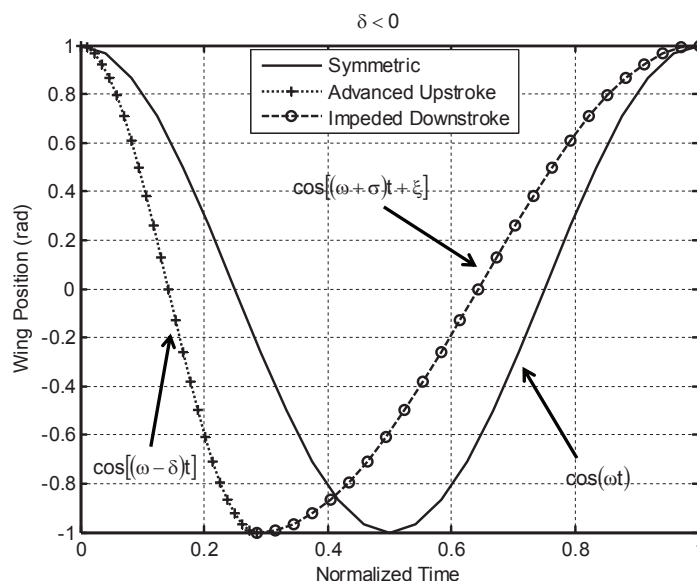


Figure 4: Wing Position for $\delta < 0$.

a large ($\frac{1}{4}$ " – 20) bolt; the puck constrained the yaw and pitch degrees of freedom. The only unconstrained degrees of freedom are the roll and horizontal translation (see Figure 8).

III. Test Article

In this section, a description of the test vehicle is provided. The vehicle consists of four-bar linkages, BLDC motors, and wings. The electronics and software to precisely control the BLDC motors are described in more detail in a later section.

The main body of the vehicle consists of a fuselage and two four-bar linkages. All parts are machined out of 6061 Aluminum. Figure 9 shows the vehicle without wings. Each four-bar linkage mechanism consists

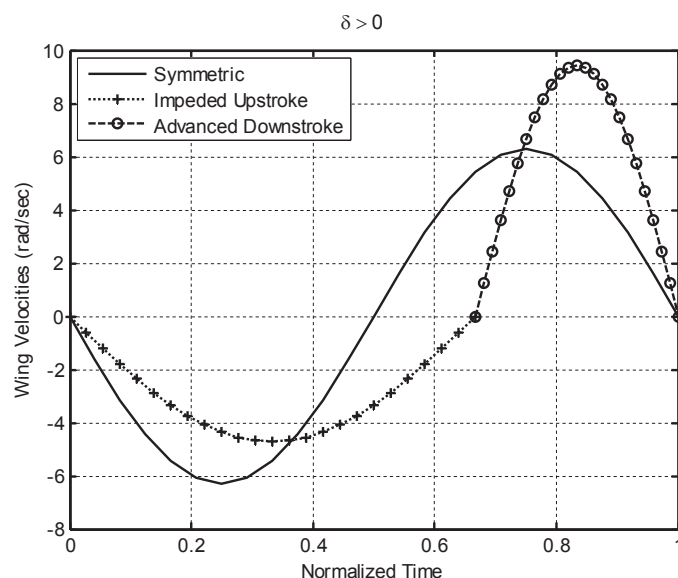


Figure 5: Wing Velocity for $\delta > 0$.

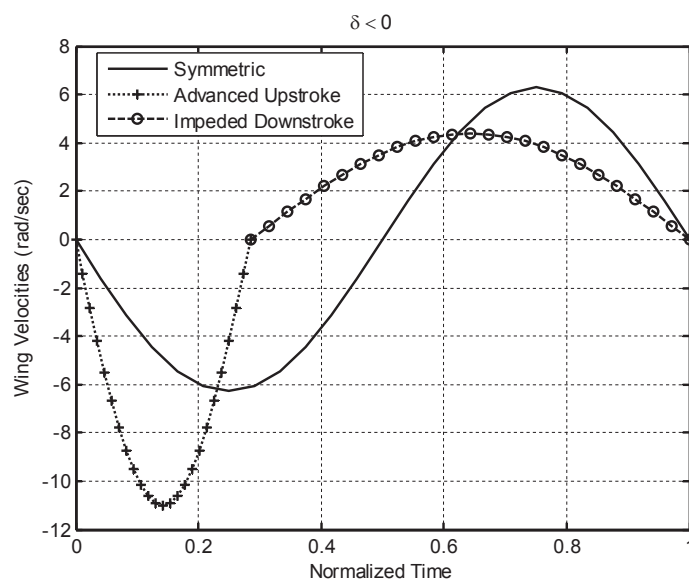


Figure 6: Wing Velocity for $\delta < 0$.

of a crank, coupler, and rocker. Custom built pins and 3 mm outside diameter ball bearings are used to provide smooth motion. The two BLDC motors are partially obscured by the fuselage. One BLDC motor connects to each crank. The fuselage and linkages weigh about 3.5 grams. Figure 10 shows a side view of the flapping wing vehicle with the motors visible on the right side. The large vertical bolt is simply used as a mount to hold the vehicle in place during testing with a force balance or on an air table.

The motors used in this work are the *Faulhaber*® 0620C006BK1674. These motors weigh about 2.5 grams. To complete the motor assembly, a 16 to 1 planetary gearbox and a magnetic encoder are included. The total weight of the motor assembly is about 5 grams. The encoder uses solid state sensors and a low inertia magnetic disk to provide two channels with a 90° phase shift and one index channel. Commutation of the motors is performed using Hall effect sensor feedback. The wings are designed to have an appropriate

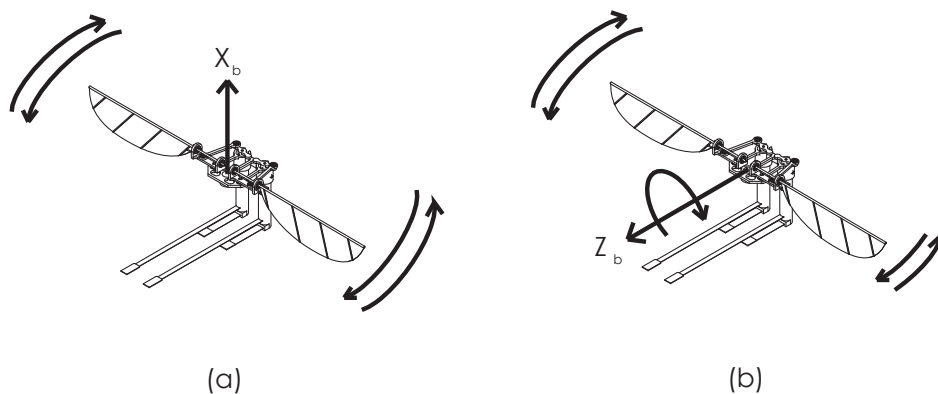


Figure 7: Degrees of Freedom Controlled by Symmetric Frequency - Vertical Translation (a) and Yaw Rotation (b).

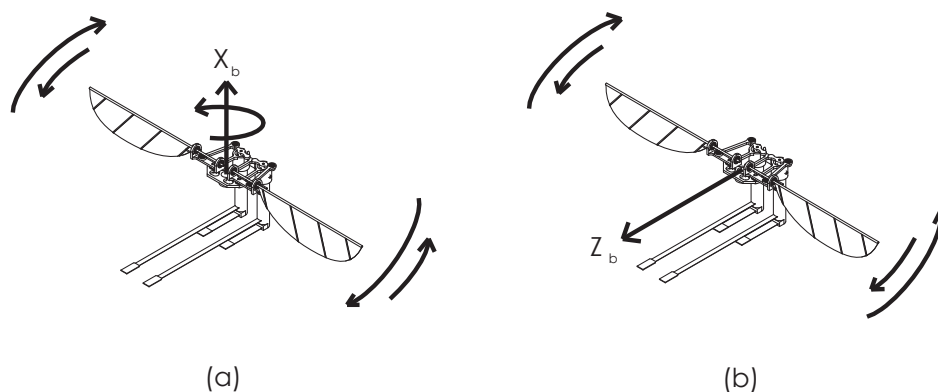


Figure 8: Degrees of Freedom Controlled by Asymmetric Frequency - Roll Rotation (a) and Horizontal Translation (b).

area moment of inertia so that a sufficient amount of cycle-averaged lift is produced. Wing design begins by developing a three-dimensional (3-D) printed positive model. A silicon mold negative is then created from the 3-D printed positive. Figure 11 shows the 3-D printed positive on the bottom and the silicon negative mold on the top. Using the silicon negative, resin is applied to 1k tow carbon fiber, which is inlaid in the mold to create the wing spar/rib structure. Capran is then adhered to the cured carbon fiber wing to form the membrane. Figure 12 shows carbon fiber being laid into the silicone model while Figure 13 shows the final wings.

IV. Remote Control Interface

The vehicle is designed to be self contained without the need for any external connections. The flapping frequencies, both symmetric and asymmetric, are provided wirelessly by a handheld transmitter. The transmitter links with a receiver to produce pulse-position-modulation (PPM) signals on the receiver output, which represent the user input to the transmitter.

The transmitter sends pulse-position-modulation (PPM) signals that are compatible with a receiver. The receiver, a *DelTang*® RX31, provides these signals to a MCU chip on the motor control printed circuit board. This receiver provides four channels in a package that weights about 0.23 grams. Figure 14 shows the *DelTang*® RX31 receiver. The MCU converts the PPM signals into symmetric and asymmetric frequencies for both the left and right motors.

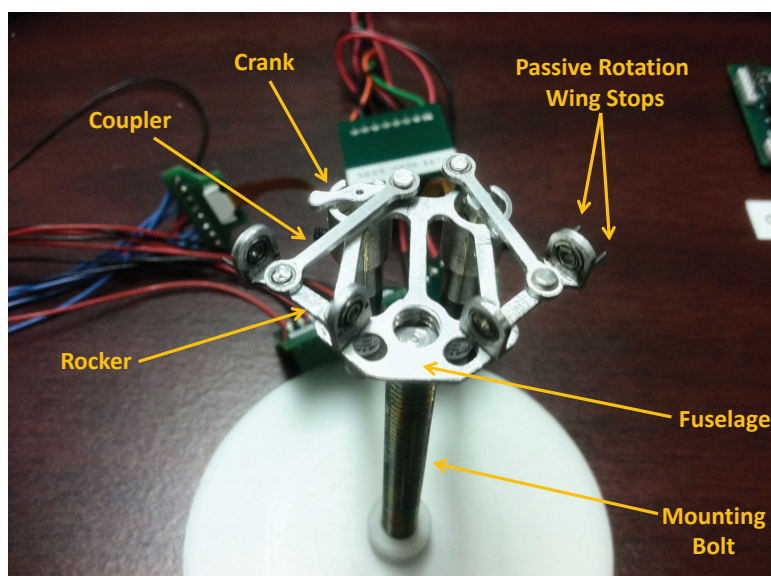


Figure 9: Fuselage and Four-Bar Linkage.

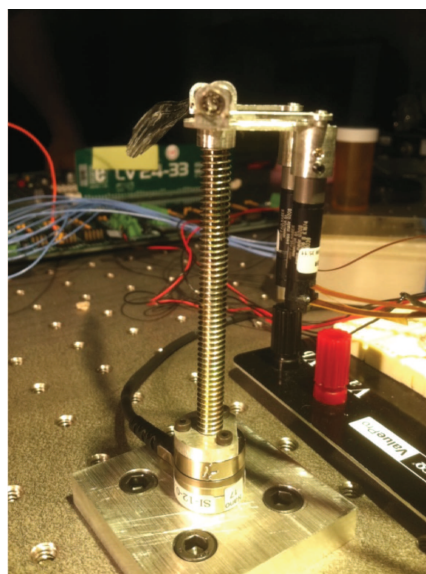


Figure 10: Side View of Flapping Wing Vehicle.

PPM signals are commonly used with radio controlled vehicles. Figure 15 shows a representative PPM signal. The signal consists of a single pulse during each phase (multiple pulses during each phase are possible). Each phase is 22 ms in duration. After the first phase, a second phase, consisting of a single (possibly different width) pulse, begins; this process continually repeats. The width of the pulses correspond to the stick settings of the handheld transmitter and have a range of 1 – 2 ms.

A single MCU is used to convert PPM signals to flapping frequencies. The MCU has built-in input capture modules which can trigger on rising and falling edges of the PPM signal. Each time the PPM signal transitions (either rising or falling), an MCU interrupt occurs and time is recorded by reading a timer value in the MCU. Once timer values for the rising edges (t_0 , t_2 , and t_4 in Figure 15) and falling edges (t_1 and t_3



Figure 11: Wing Positive (bottom) and Negative (top) Molds.

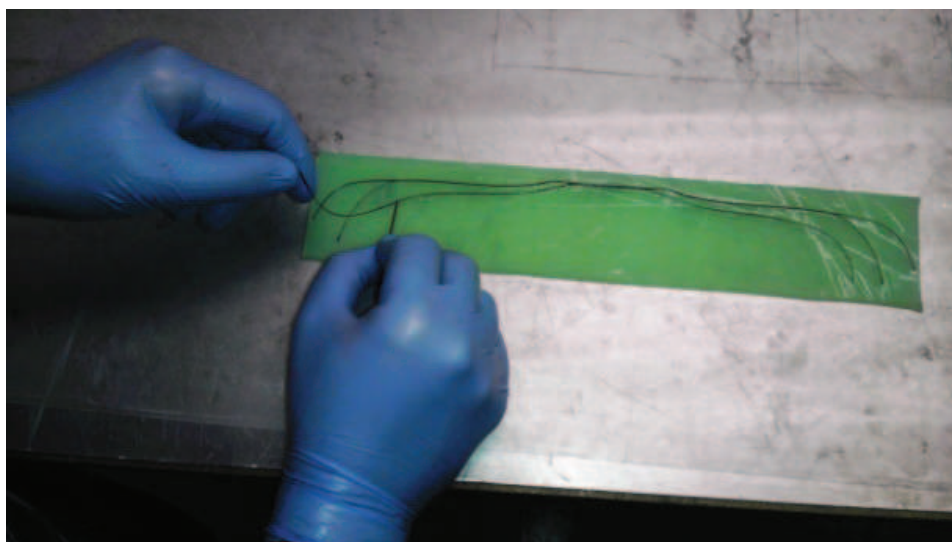


Figure 12: Wing Spar and Rib Construction Using Wetted Carbon Fiber.

in Figure 15) have been recorded, the pulse width is determined by simply taking a difference:

$$\Delta t_i = t_i - t_{i-1} \quad i = 1, 3, 5, \dots \quad (3)$$

These pulse width values are then mapped to frequencies. Once the upstroke and downstroke frequencies are obtained, they are converted to time durations for each stroke. The time durations are used directly by the motor commutation MCUs. The current experimental setup has four of these operations taking place at all times. Two are for the left and right symmetric frequency and two are for the left and right asymmetric (split-cycle) frequency. The transmitter has two control sticks with two degrees of freedom each. The left vertical stick position maps to symmetric frequency for both the left and right wings, specifically, upward tilting increases the symmetric frequency of both wings equally. The left horizontal stick position maps to differential symmetric frequency between the left and right wings, specifically, tilting the stick to the right increases the symmetric frequency of the left wing as compared to the right wing. The right vertical stick position maps to asymmetric frequency of both wings, with upward tilt increasing the velocity of the wings

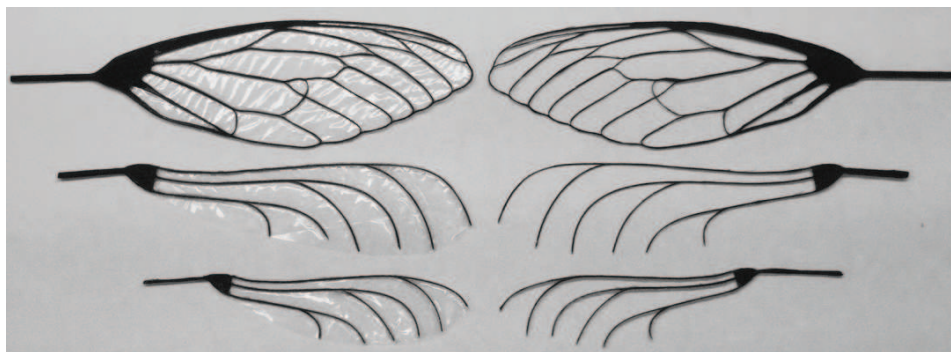


Figure 13: Wings With Membrane (left) and Without Membrane (right).

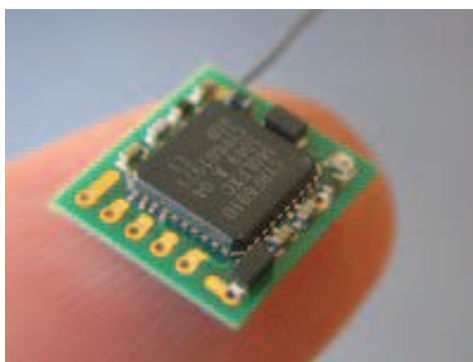


Figure 14: *DelTang*® RX31 Receiver.

on the upstroke relative to the downstroke. The right horizontal stick position maps to asymmetric frequency differential between the left and right wings, specifically, tilting to the right causes the downstroke of the left wing to slow compared to its upstroke while the upstroke of the right wing slows as compared to its downstroke. The PPM signals are constantly monitored by the MCU, however, the resulting information is only used at the completion of a full wingbeat cycle. Hence, the symmetric and split-cycle frequencies are only allowed to change once per full wingbeat cycle, in particular, at the beginning of a cycle.

V. Motor Control

The motor controller's objective is to produce a cosinusoidal wing stroke motion with the ability to control the split-cycle characteristic of the motion, namely to traverse the upstroke and downstroke at different velocities. This is achieved by employing an encoder to detect the crank position and velocity, then performing trajectory tracking with proportional, derivative, and integral (PID) control. The reference trajectory is generated by a time indexed lookup-table based on the inverse kinematics of the crank-rocker four-bar linkage mechanism applied to a cosinusoidal rocker position reference.

Figure 16 provides an overview of the control actuation. The encoder provides incremental position measurements via a counter. A timer is used to measure the duration between encoder transitions, which is

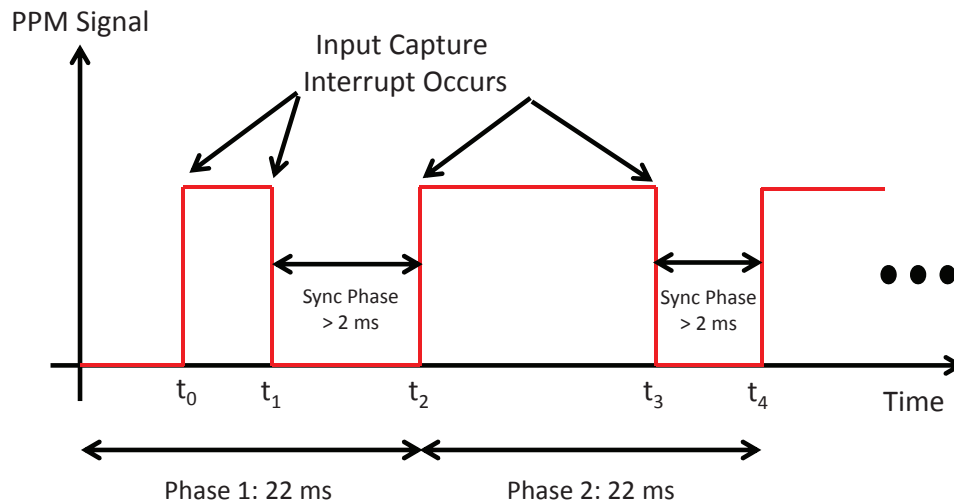


Figure 15: Pulse Position Modulation Signal.

converted into a velocity measurement. Timer information is also used for cumulative numerical integration of the position error. The measured position and velocity are compared to reference values generated using a period timer and lookup-tables. The period timer is set according to the commanded lengths of time between table entries, and the increment values may be different for each half of the two tables in order to generate a split-cycle reference signal. The position and velocity errors are computed and the position error must be wrapped due to the periodic nature of the system. Gains are applied to the PID errors and the resultant PID signal is fed to a pulse width modulation (PWM) generator to set the duty-cycle of the motor energization coil. Hall effect sensors provide feedback to control which motor coils are energized in order to commute the motor in the direction indicated by the sign of the PID control signal.

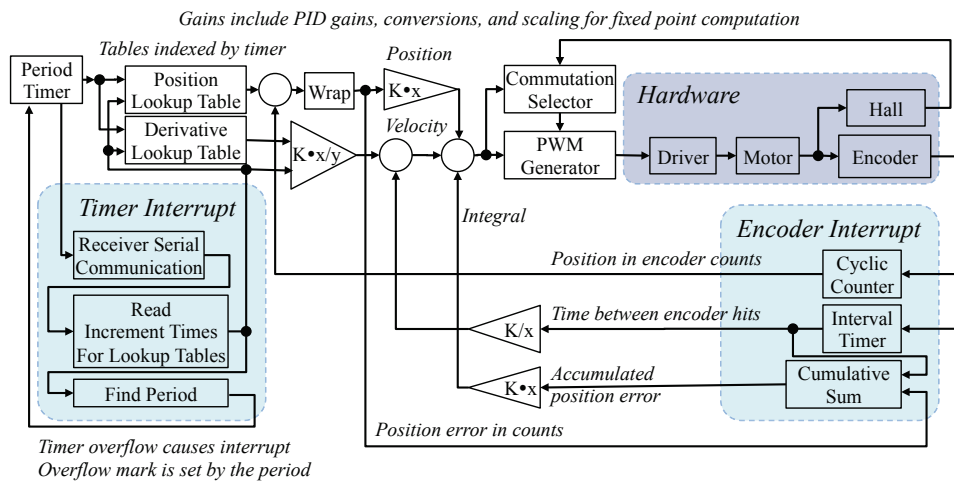


Figure 16: Block Diagram Overview of the Control Actuation.

A. Kinematic Inversion

A four-bar linkage in a crank-rocker configuration is used to transform the rotational motion produced by a BLDC motor and planetary gearbox into reciprocating wing root motion. In general, the time that it takes for a rocker to rotate counter-clockwise differs from the time that it takes to rotate clockwise over the course of one constant velocity crank rotation. Due to the nonlinear kinematics of the four-bar linkage

motion translation, a constant crank velocity yields a non-symmetric rocker position profile (the upstroke and downstroke wing positions are not symmetric and their velocities are not equal and opposite in magnitude). In order to produce controlled split-cycle motion without physically altering the geometry of the four-bar mechanism, the crank velocity must be precisely controlled. Thus, in order for a four-bar mechanism to produce temporally symmetric motion over the course of a single crank rotation, the crank velocity must be varied in a specific manner. The problem of computing the crank velocity required to produce a desired rocker motion profile is termed kinematic inversion. This process can be performed offline and implemented, using precomputed tables, in real-time.

It can be shown⁵ that the relationship between the rocker angle and the crank angle in a four-bar linkage can be written as

$$\phi_l = 2 \arctan 2 \left(\frac{-K_1(\theta_l) \pm \sqrt{K_1(\theta_l)^2 - K_3(\theta_l)^2 + K_2(\theta_l)^2}}{K_3(\theta_l) - K_2(\theta_l)} \right) \quad (4)$$

where ϕ_l is the rocker angle, θ_l is the crank angle, $\arctan 2$ denotes a 4 quadrant arctangent function that is dependent upon the signs of the numerator and denominator and,

$$K_1(\theta_l) = -2l_1l_3 \sin \theta_l \quad (5)$$

$$K_2(\theta_l) = 2l_3(l_o - l_1 \cos \theta_l) \quad (6)$$

$$K_3(\theta_l) = l_o^2 + l_1^2 - l_2^2 + l_3^2 - 2l_o l_1 \cos \theta_l \quad (7)$$

where l_o denotes the distance between the line of centers, l_1 denotes the crank length, l_2 denotes the coupler bar length, and l_3 is the rocker length. Figure 17 shows the geometry of the four-bar mechanism. To

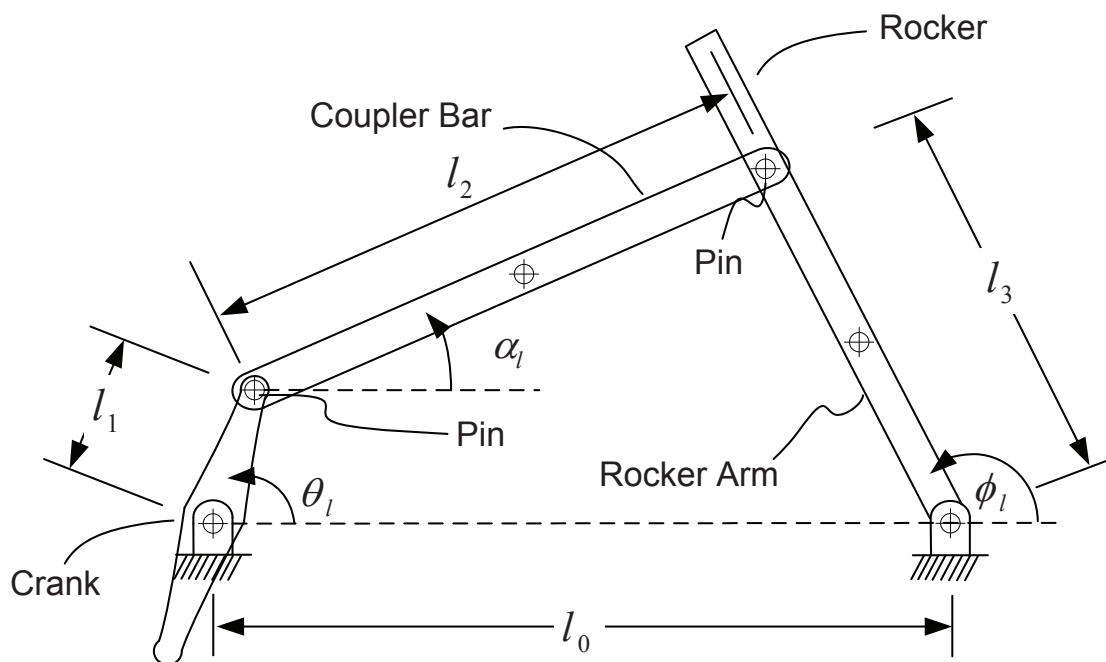


Figure 17: Four-Bar Linkage Geometry.

illustrate the temporally asymmetric behavior of a rocker arm linked to a constant velocity crank, consider the specific case described in Table 1. Figure 18 shows that the rocker arm angular velocity oscillates and exhibits temporally asymmetric behavior across each wingbeat cycle when driven by a constant velocity crank. In particular, the rocker arm and thus the wing spar achieve higher stroke plane angular velocity on the downstroke than on the upstroke. The dynamic pressure on the wing will therefore be higher on the downstroke than on the upstroke, which will lead to a finite value of cycle-averaged horizontal force when the drag profile is resolved into the body-axis system. For symmetric flapping, the upstroke and downstroke

angular velocity profiles should be equal in magnitude and opposite in sign. In this way, the cycle-averaged horizontal force will be zero. In order to achieve symmetric flapping, kinematic inversion must be performed.

Description	Variable	Value	Unit
Line-of-centers length	l_0	1.936	cm
Crank length	l_1	0.461	cm
Coupler Bar pin-to-pin length	l_2	2.060	cm
Rocker pin-to-pin length	l_3	0.670	cm

Table 1: Parameters for the Four-Bar Linkage.

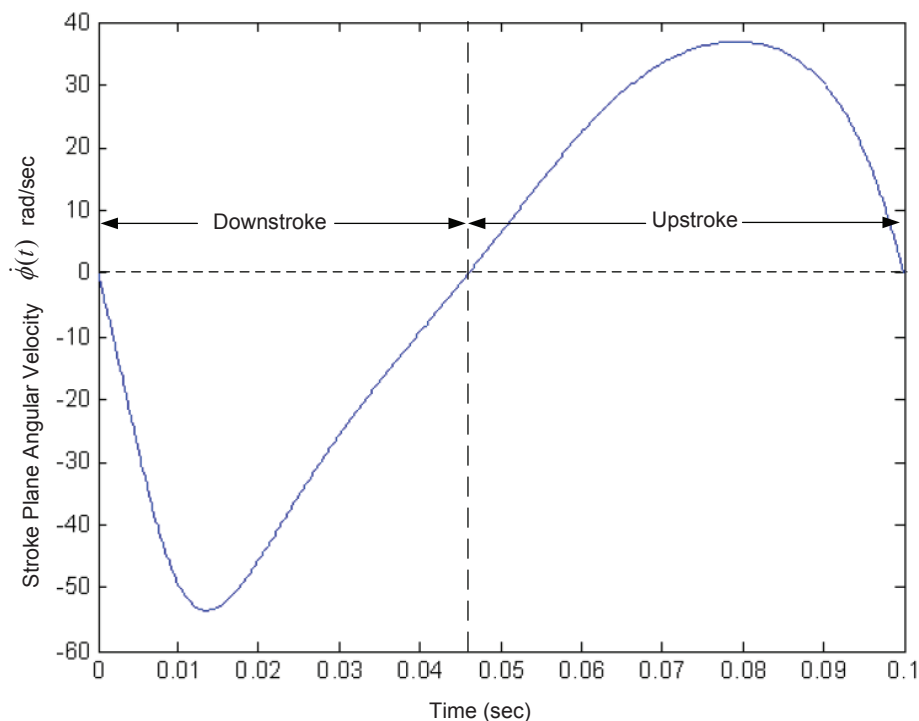


Figure 18: Wing Angular Velocity Profile.

The inverse kinematics can be derived either analytically or numerically from the forward kinematics. The inverse kinematics computations are performed offline to ease computational load, trading computational operations for memory required to store offline generated lookup-tables. For a full revolution of the crank, the rocker position spans the same sector of angular positions twice but with opposite signs on the rocker velocity, i.e., the upstroke and the downstroke. Thus, the inverse kinematics must either be computed separately for the upstroke and downstroke, or by explicitly including the sign of rocker velocity, in order to create a continuous one-to-one mapping from the rocker to the crank.

The position lookup-table is produced by first sampling a cosinusoidal rocker position waveform with a normalized period of 1, uniformly in time. The cosinusoidal rocker position waveform is required to yield symmetric flapping. The inverse kinematics are applied to the samples of the rocker position to produce a corresponding table of crank positions. In other words, given the desired cosinusoidal rocker position waveform, ϕ_l , Eq. 4 is used to compute the crank angle, θ_l , at each sampled point in the desired ϕ_l trajectory. This resulting crank position table is the position lookup-table used to generate the position reference signal. A table of the center differentials is also computed for each index i by taking the difference between the $(i+1)$ and $(i-1)$ entries of the position lookup-table. The central difference table is used to compute the velocity

reference lookup-table. The end points of the table are special cases to account for the periodicity and continuity of the waveform. Figure 19 illustrates the lookup-table generation for a 32 entry table, indicated by x symbols on the plots. In practical implementation, 1024 entries were used. Note that the position table is non-linear as a result of the inverse-kinematics requiring a non-constant crank velocity to produce a cosinusoidal rocker motion.

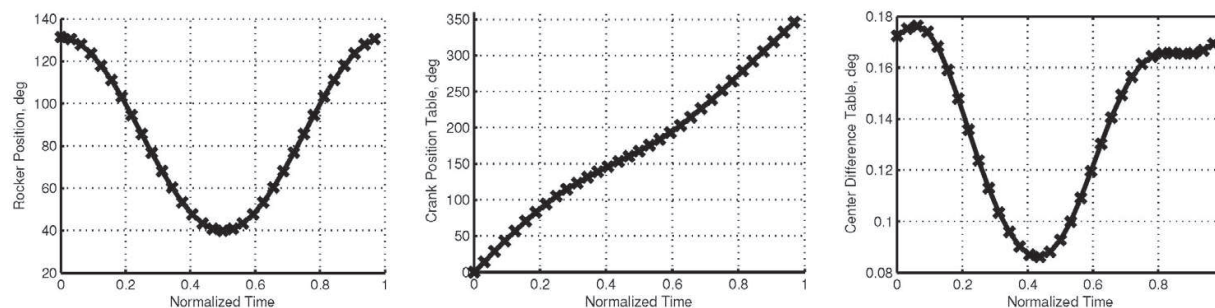


Figure 19: Reference Rocker Position, Crank Position to Produce Reference Rocker Position, and Center Difference Lookup Table.

B. BLDC Motor Control Software and Hardware

Figure 1 shows a block diagram of the system used to control the motion of the flapping wings. Working from left to right, a conventional RC aircraft transmitter is used to wirelessly send signals to the *DelTang*[®] RF receiver and custom designed circuit board that contains a network of microcontrollers (MCUs) and amplifiers that reside on the aircraft. The circuit board makes use of surface mount components that provide the necessary capabilities in a package of suitable size, weight, and power. RC aircraft use a pulse-position-modulated communication protocol to communicate operator intent through a receiver onboard the aircraft. In order to perform remote controlled SCCPFM on the stroke of both wings, user commands are mapped from stick positions on the RF transmitter, to PPM commands generated by the RF receiver, and then to symmetric and asymmetric (split-cycle) wingbeat frequency commands computed by the MCU connected to the receiver. The frequency commands are translated on the same MCU into timing commands transmitted via serial communication to two MCUs dedicated to controlling each of the wing actuation motors. The motor control MCUs monitor the timing and position of the wings, requesting new timing commands at the end of each wingbeat, effectively giving the user control of the velocity profile of each stroke. The motor control MCUs also compute crank time and position targets for the current wingbeat cycle while monitoring the motor position sensors to provide pulse-width-modulation (PWM) based current control to each motor. The control responds to differences between the target and actual motor positions and uses the commutation sequence of the BLDC motors according to Hall effect sensor feedback. The target motor positions come directly from tables generated by the inversion of the kinematics relating the motor position to the wing position. The final element in each motor control system is a current amplifier that consists of field-effect-transistors arranged in an H-bridge configuration. Each current amplifier is wired directly to the phases of each BLDC motor.

Figures 20 and 21 show the custom built motor control hardware through four stages of development. Stage 1 used large development boards to prove out the concept. Following stage 1, three additional versions of custom development boards were built. The stage 2 version separated the MCU and amplifier portions of the boards. Stage 3 combined the MCU and amplifier circuits into one self contained board, while stage 4 is operationally identical to stage 3, except that it is a flight weight board that eliminated all circuit overload and reverse polarity protection and optimized components for size and weight. The stage 1, 2, and 3 hardware are single sided boards, while the stage 4 hardware is a two sided board. Figure 21 shows both sides of the stage 4 board. These boards connect to the BLDC motors and drive the motors to achieve the commanded wingbeat velocity trajectory. Each of the control and amplifier boards are reprogrammable. The motor control and PPM to frequency conversion software are downloaded to the three MCUs on the boards to provide the desired wing motion. Table 2 shows the components used in the stage 4 control and

driver board. Additional thru holes and pads for connecting to the board are present in the actual design but not included in the component table.

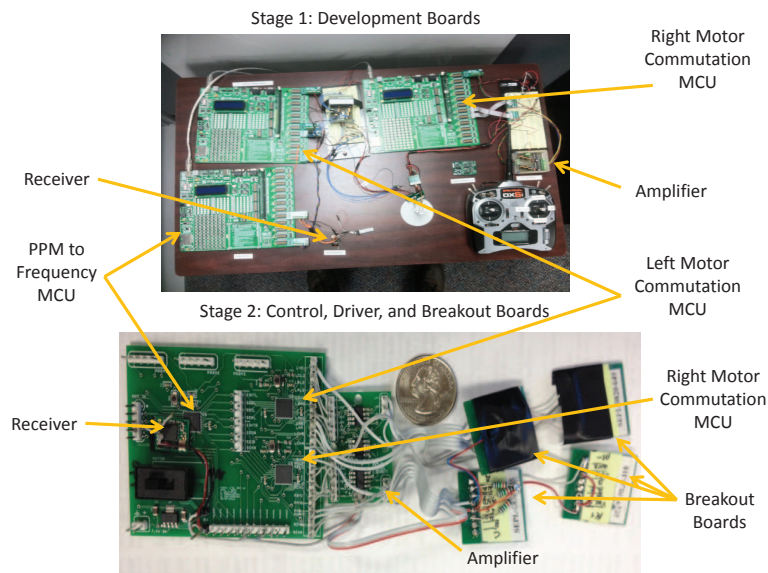


Figure 20: Stage 1 and 2 Motor Control Boards.

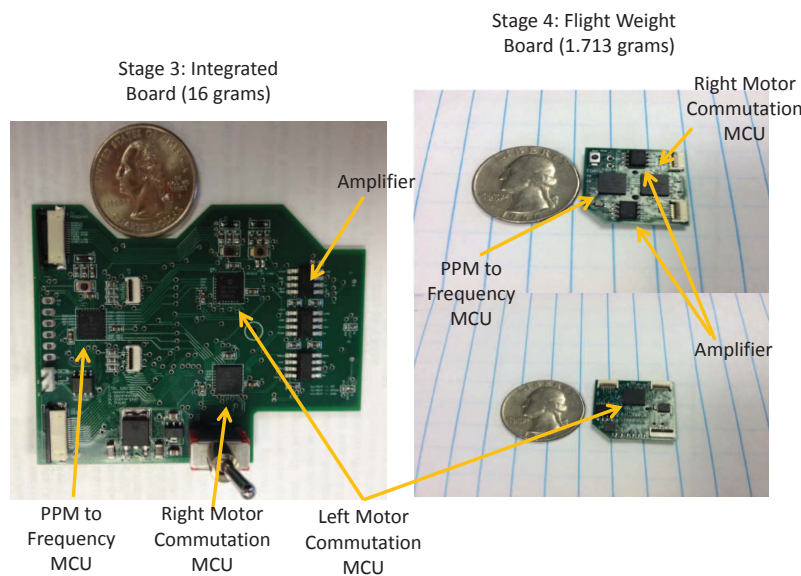


Figure 21: Stage 3 and 4 Motor Control Boards.

C. Closed Loop Wing Trajectory Tracking

The closed loop wing trajectory tracking uses proportional-integral-derivative (PID) control to force the measured crank position to track the crank position reference. The combined sum of the PID control output signal commands the duty-cycle of the BLDC motor commutation. As described above, the crank position and derivative reference are generated using a timer indexed lookup-table. The proportional and derivative terms are obtained by multiplying the difference between the position and velocity measurements and the reference. The position and velocity are measured using an encoder. The integral term is updated at

Part Number	Description	Package
PIC24FJ64GA102-I/ML	MCU	28-QFN
LP38690SD-5.0/NOPB	Voltage Regulator	6-LLP
LP38690SD-3.3/NOPB	Voltage Regulator	6-LLP
ZXMHC6A07N8TC	H-Bridge	8-SOIC
B3U-1000P	Switch	SMT
503480-1000	10 Position Connector for MCU Programming	SMT
XF2W-0815-1A	8 Position Connector for Motor and Encoder Connections	SMT
ERJ-1GEF1002C	10k Ohm Resistor	SMT
ERJ-1GEF1501C	1.5k Ohm Resistor	SMT
ERJ-1GEF4700C	470 Ohm Resistor	SMT
C0603X5R1A104K	0.1 μ F Capacitor	SMT
C1608X5R1A105K	1 μ F Capacitor	SMT
C0603X5R0J105M	1 μ F Capacitor	SMT
21020-0101	Flat Flexible Cable for MCU Programming	N/A

Table 2: Components Used in Stage Four Control and Driver Board.

every encoder transition as a cumulative sum of products of position errors and duration between encoder transitions, forming a non-uniform discrete backwards Euler integration of the position error. The PID gains were assigned according to LQR tuning techniques applied to a simplified motor and controller model, then manually tuned to compensate for modeling error.

The crank position reference is generated using the position lookup table and a resettable timer. At the start of each wingbeat cycle, a timer with a period equal to the stroke period is started and used for lookup-table indexing. While indexing the first half of the table (downstroke), the index is incremented by 1 after each Δt_d seconds increment of the timer, and during the second half of the table (upstroke), the index is incremented after each Δt_u seconds increment of the timer. At the end of the period, the motor control MCU requests new commanded Δt_u and Δt_d from the receiver MCU. The receiver MCU obtained the commands from the transmitter, i.e., they are assigned by the remote control user.

The symmetric and split-cycle frequency are set by adjusting the Δt_u and Δt_d values. For an N element lookup-table the procedure is as follows: to produce a stroke with a period T , assign $\Delta t_u = \Delta t_d = T/N$. To produce a wingbeat with non-zero split-cycle, such that the duration of the upstroke is $T_u/2$ and the downstroke is $T_d/2$, assign $\Delta t_u = T_u/N$ and $\Delta t_d = T_d/N$.

The position reference at any given time is obtained by using the corresponding value of the lookup-table index to obtain the value at that index from the crank position table. The velocity reference is obtained by first using the same index to look up the corresponding entry at the same index from the center differential table. The entry is then divided by $2\Delta t_u$, if the index is in the first half of the table, or $2\Delta t_d$ if it is in the second half. This computation is a standard numerical derivative and yields the velocity reference.

The position and velocity of the crank are measured using an encoder. The rotor of the motor is equipped with two incremental encoders, positioned 90° out of phase. The 64 slot encoders are configured to report every 4th transition and the rotor is connected to the crank via a 16:1 gear ratio, thus there are $64 * \frac{1}{4} * 2 * 2 * 16 = 1024$ transitions for every full crank rotation, i.e, the crank position measurement resolution is $2\pi/1024$ rad. Standard techniques can be used to deduce relative position and direction of rotation using appropriate encoder state transition progression assessment and counting. A timer is used to measure the time duration between encoder transitions, Δt , thus yielding a discrete velocity measurement of $2\pi/(1024\Delta t)$ rad/sec for every encoder transition. If the encoder status progression indicates a reversal in rotational direction, because of system dynamics or disturbance, the velocity measurement is ignored, since it is impossible to know the rotational distance traveled between the current and previous encoder transitions.

The motor control MCUs are equipped with pulse-width-modulation (PWM) modules. These modules are employed to produce the duty-cycle commanded by the PID control signal. There are three modules

used for each motor and they command the state of the lower side of six power metal oxide semiconductor field effect transistors in a half H-bridge configuration, while three other pins of the MCU control the high side. Each motor is equipped with three Hall effect sensors to detect the position of the motor rotor. The PID control signal is computed as often as possible on the motor control MCUs and each time the Hall effect sensors are read. Their status and the sign of the PID control signal dictate which module is assigned the commanded duty-cycle and which high side pin is used as its complement, in accordance to the commutation sequence for the BLDC motors. Thus, the desired crank actuation is performed in accordance to the wing trajectory tracking controller software.

VI. Qualitative Motion on Air Table

The air table provides the lift equal to weight condition and allows isolation of the degrees of freedom that can be controlled with split-cycle, namely, horizontal translation and roll. Figure 22 show the results of a test of the split-cycle control law on an air table. The first column of the figure shows the commanded stick position on the RF transmitter. For example, the upper left hand corner shows the stick pulled back, which commands the vehicle to translate aft. The rows of six pictures, shown to the right of each stick position, show the time elapsed motion of the vehicle. The first picture in each row has an arrow to show the desired direction of translation or rotation. The amount of time elapsed during each set of six frames is 5 seconds. In the first row, the vehicle starts near the top of the air table and moves backward, while exhibiting almost zero roll. The second row of plots shows the stick to the left, which commands a counter-clockwise rotation of the vehicle. The six frames in the second row show the vehicle rotating counter clockwise. The third row has the stick to the right, which commands a clockwise rotation of the vehicle, and the plots to the right show the vehicle rotating with a limited amount of translation. In the last row, the stick is forward, which commands a forward translation of the vehicle. The six frames to the right show the vehicle moving forward with nearly zero rotation.

VII. Conclusions

This paper discussed the construction, control, and testing of a flapping wing vehicle. The vehicle utilizes the split-cycle control strategy to provide horizontal translational and roll control. An experiment, using an air table, was conducted to verify the efficacy of split-cycle control. The air table provides the lift equal to weight condition and allows isolation of the degrees of freedom that can be controlled with split-cycle, namely, horizontal translation and roll. The results of this experiment were that the vehicle could independently translate, in the horizontal direction, and roll as commanded by an operator. The experimental evidence supports the hypothesis that split-cycle constant-period frequency modulation is a viable method to produce roll and horizontal translation in a flapping wing vehicle. Furthermore, the results suggest that four degrees of freedom could be directly controlled by using two physical actuators if the technique were applied to a free-flight vehicle rather than a vehicle constrained to maneuver in the horizontal plane.

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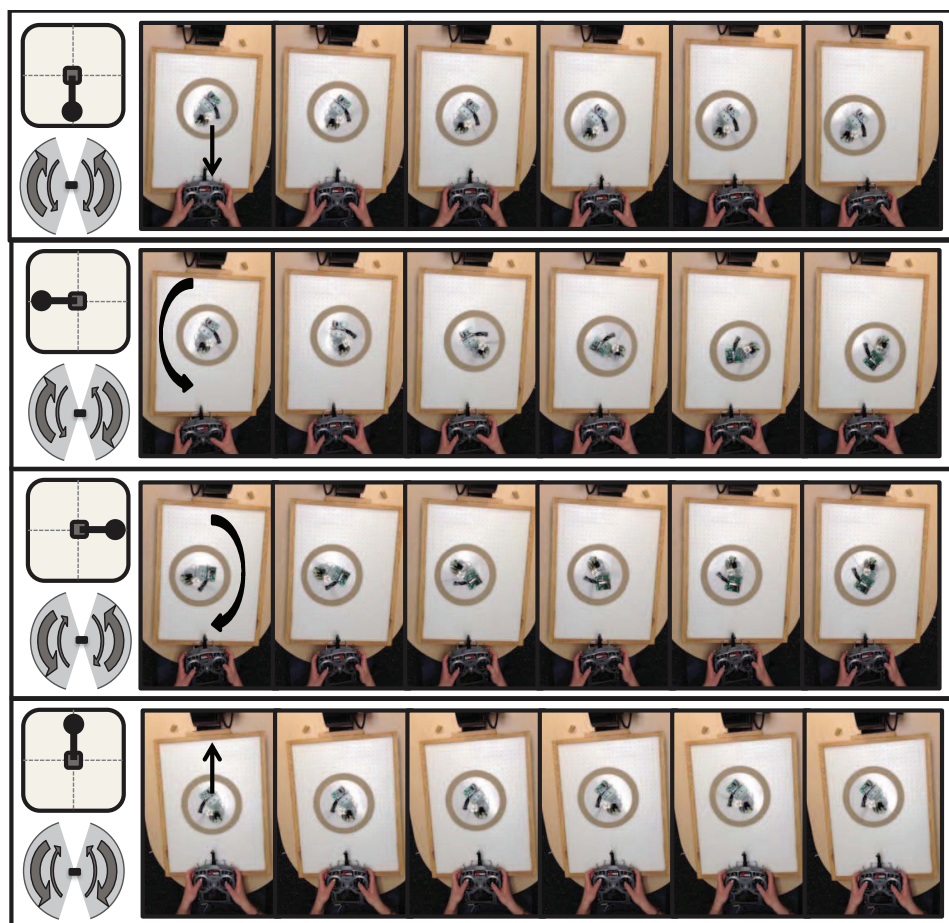


Figure 22: Air Table Experiment Results.